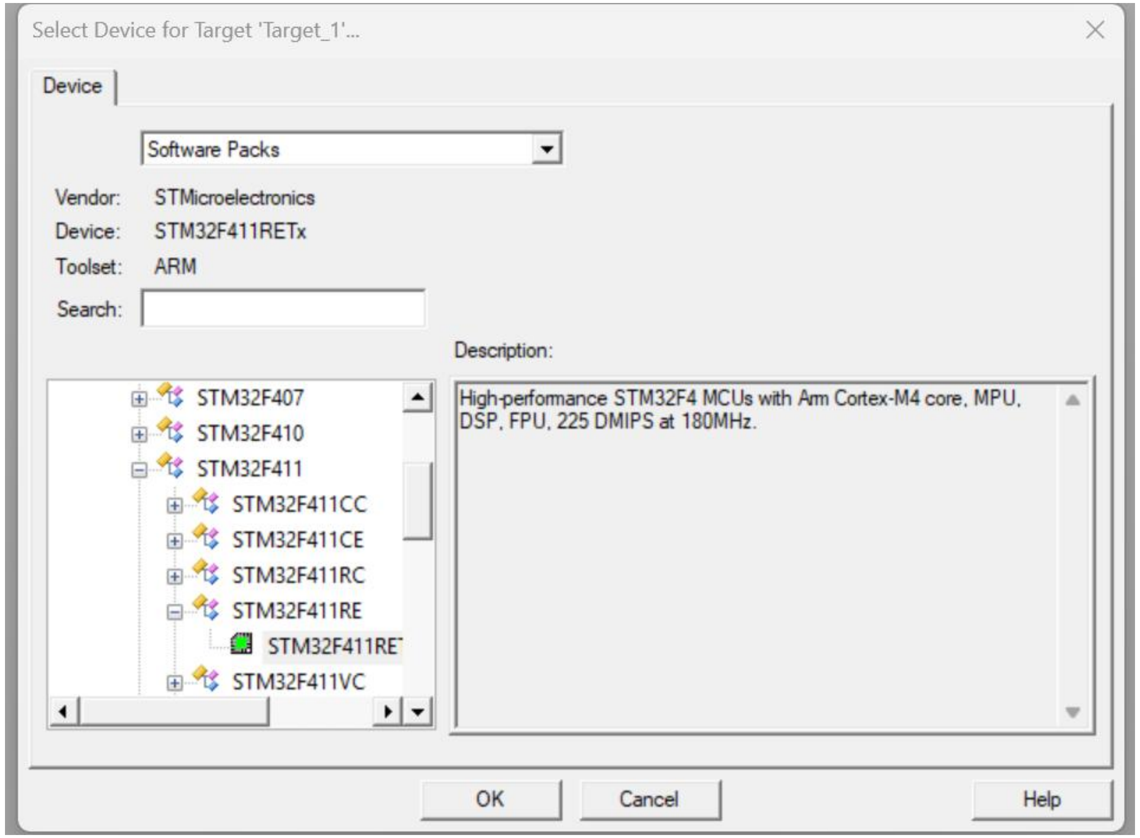
Nucleo-F411RE

Phase 1: Create the Folder and Project

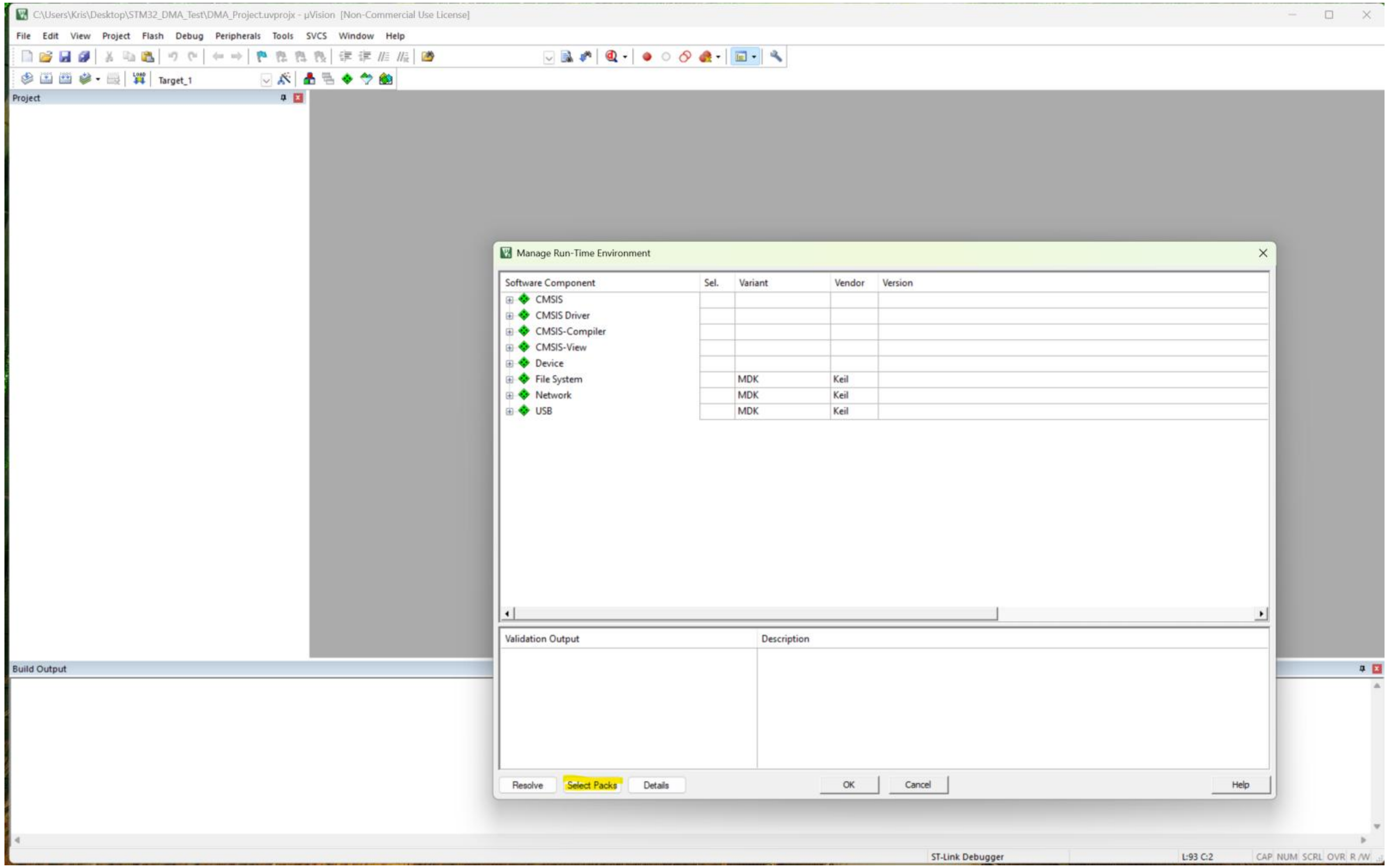
1. **Create a New Folder:** In Windows, create a folder named `STM32_DMA_Test`. (Always keep your projects in their own folders to avoid file mix-ups).
2. **Open Keil:** Go to **Project -> New uVision Project...**
3. **Save it:** Navigate to your new folder and name the project `DMA_Project`.
4. **Select Device:** In the search box, type `STM32F411RE`. Select it and click **OK**.

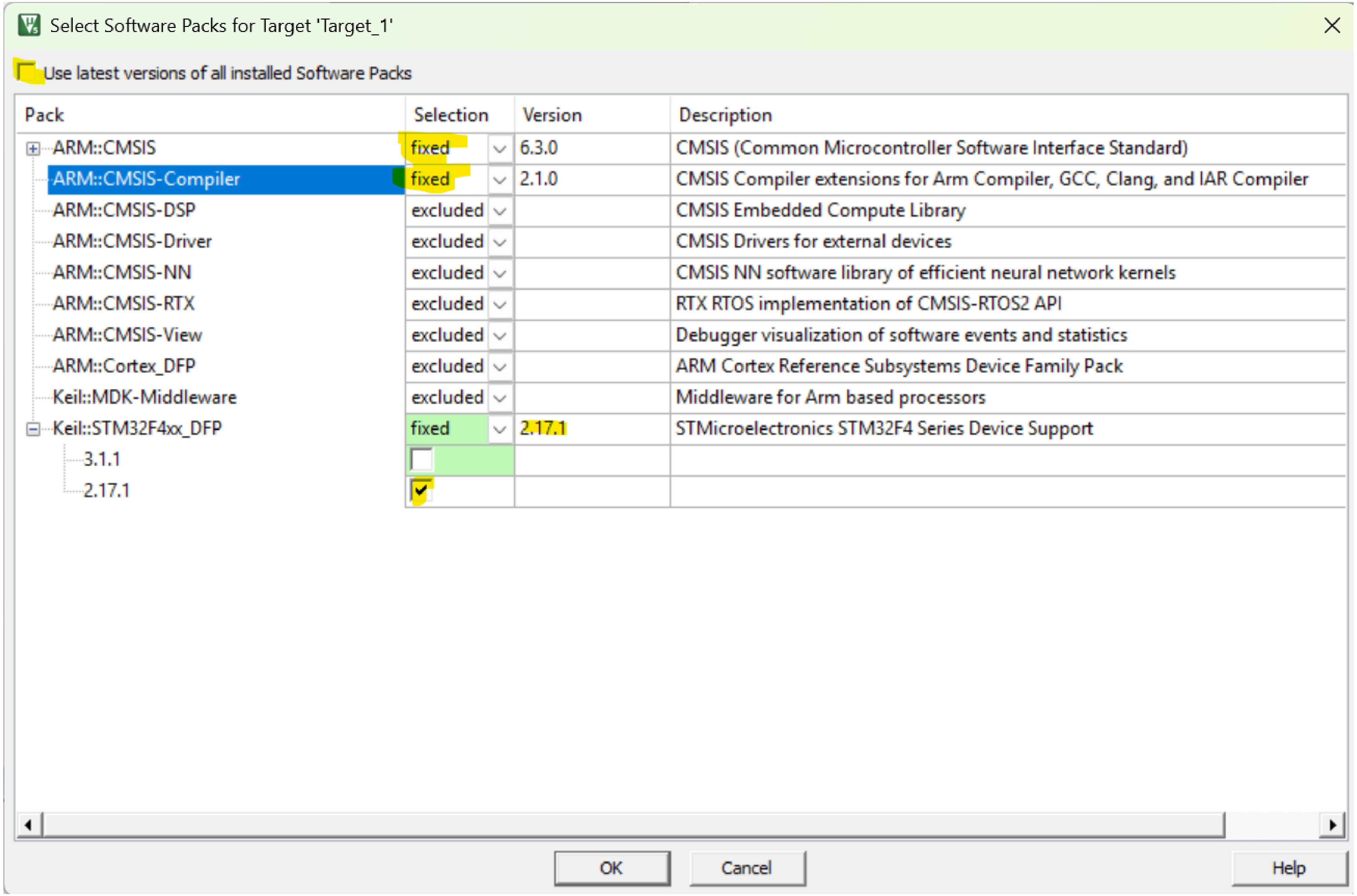


Phase 2: Force the "Classic" Pack (Crucial Step)

A window titled **"Manage Run-Time Environment"** will open. **Do not check anything yet.**

1. Click the **Select Packs** button at the bottom left.
2. **Uncheck** the box at the top: *"Use latest versions of all installed Software Packs"*.
3. Find **Keil::STM32F4xx_DFP** in the list.
4. Click the version number (likely 3.1.1) and change it to **2.17.1**.
5. Click **OK**.

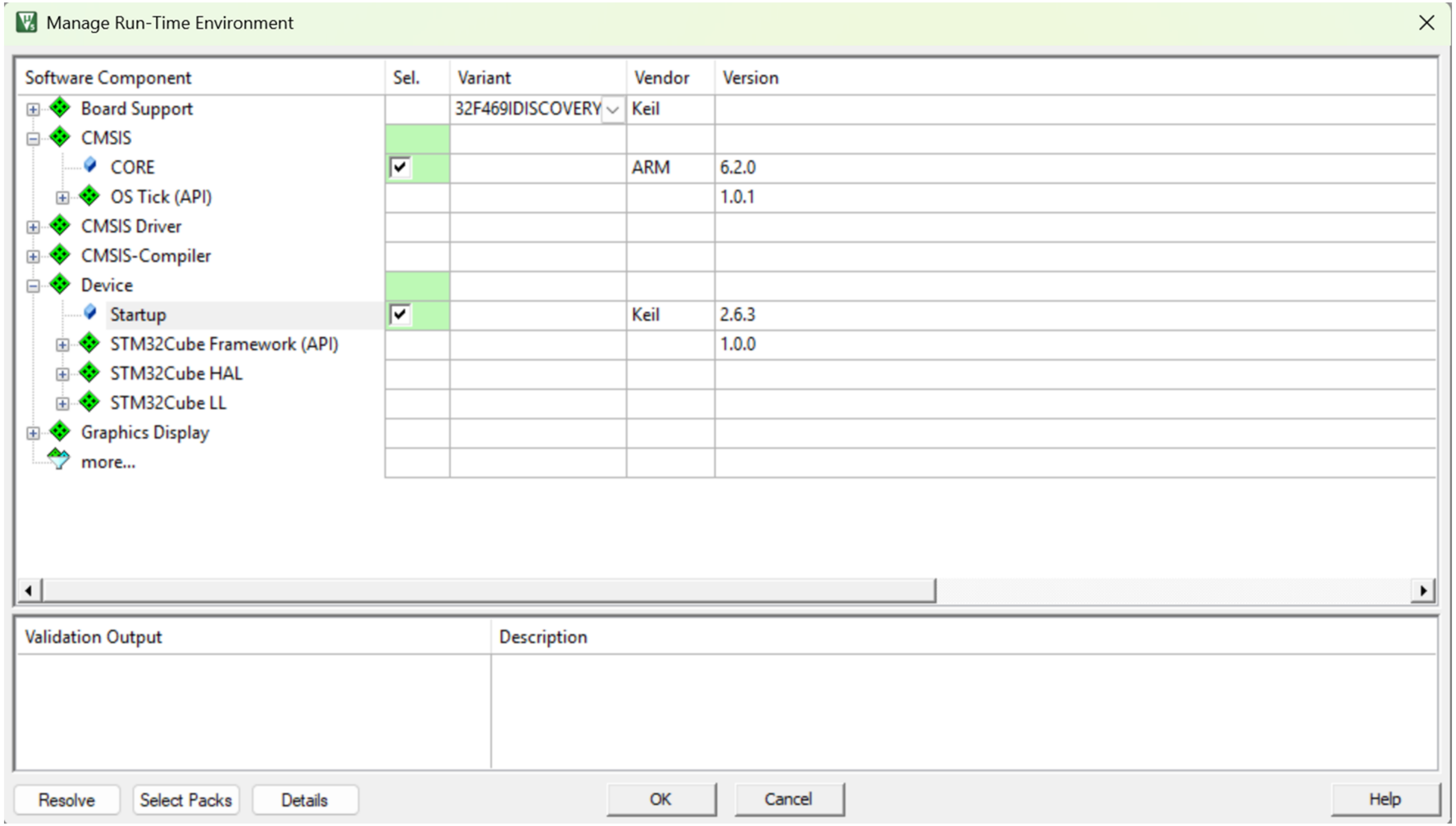



Select Software Packs for Target 'Target_1'
Use latest versions of all installed Software Packs
Pack
Selection
Version
Description
ARM::CMSIS
fixed
6.3.0
CMSIS (Common Microcontroller Software Interface Standard)
ARM::CMSIS-Compiler
fixed
2.1.0
CMSIS Compiler extensions for Arm Compiler, GCC, Clang, and IAR Compiler
ARM::CMSIS-DSP
excluded
CMSIS Embedded Compute Library
ARM::CMSIS-Driver
excluded
CMSIS Drivers for external devices
ARM::CMSIS-NN
excluded
CMSIS NN software library of efficient neural network kernels
ARM::CMSIS-RTX
excluded
RTX RTOS implementation of CMSIS-RTOS2 API
ARM::CMSIS-View
excluded
Debugger visualization of software events and statistics
ARM::Cortex_DFP
excluded
ARM Cortex Reference Subsystems Device Family Pack
Keil::MDK-Middleware
excluded
Middleware for Arm based processors
Keil::STM32F4xx_DFP
fixed
2.17.1
STMicroelectronics STM32F4 Series Device Support
3.1.1
2.17.1
OK
Cancel
Help

Phase 3: Select the Components

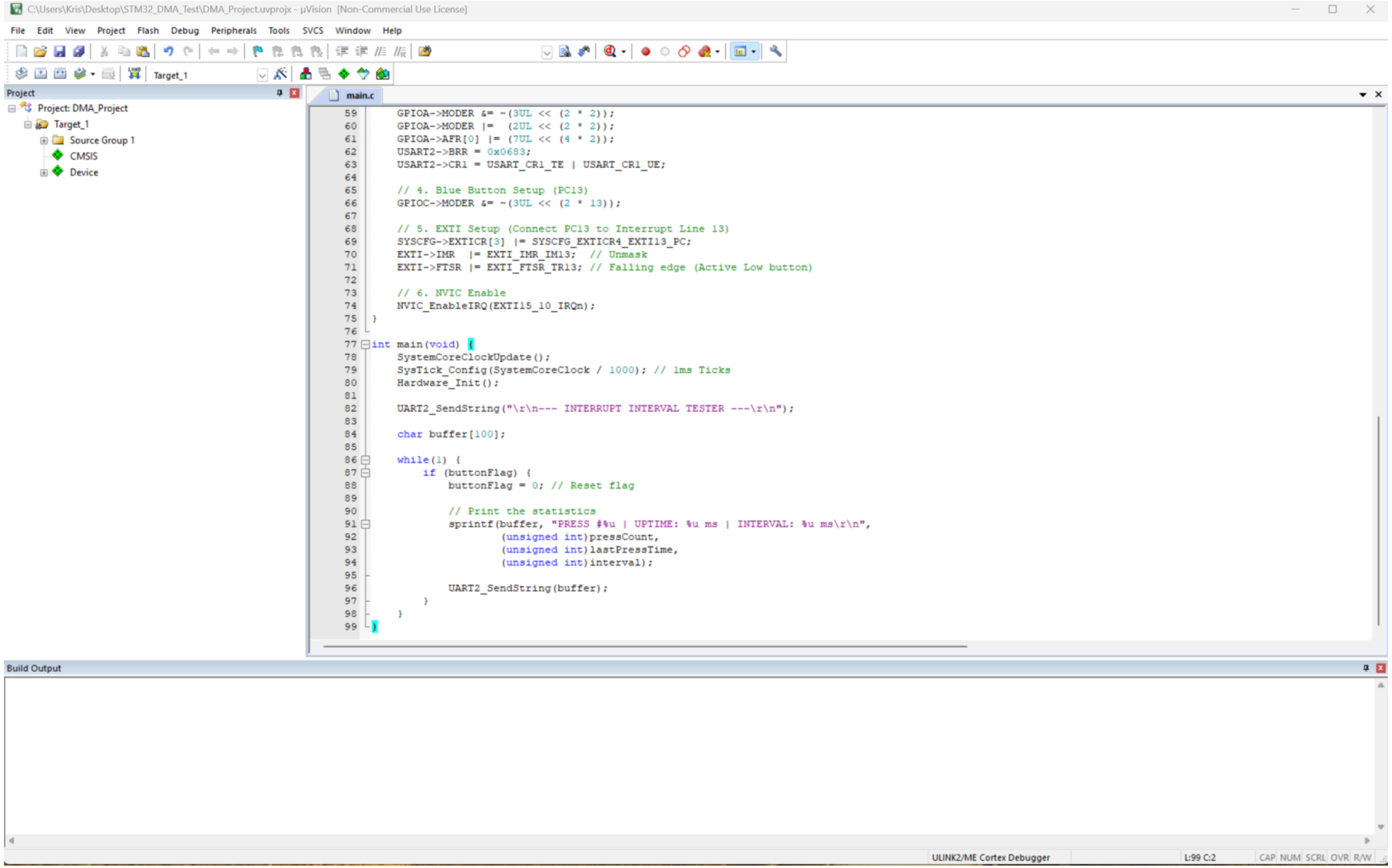
Once you click OK in Phase 2, you'll be back in the **Manage Run-Time Environment** window.

1. Expand **CMSIS** and check **CORE**.
2. Expand **Device** and check **Startup**.
    1. *Note: If you see "STM32Cube" options, ignore them. You only want the plain "Startup" box.*
3. Click **OK**.



Phase 4: Create the Code File

1. In the **Project** window on the left, right-click **Source Group 1**.
2. Select **Add New Item to Group 'Source Group 1'...**
3. Choose **C File (.c)**, name it `main`, and click **Add**.
4. Paste your code into the editor.

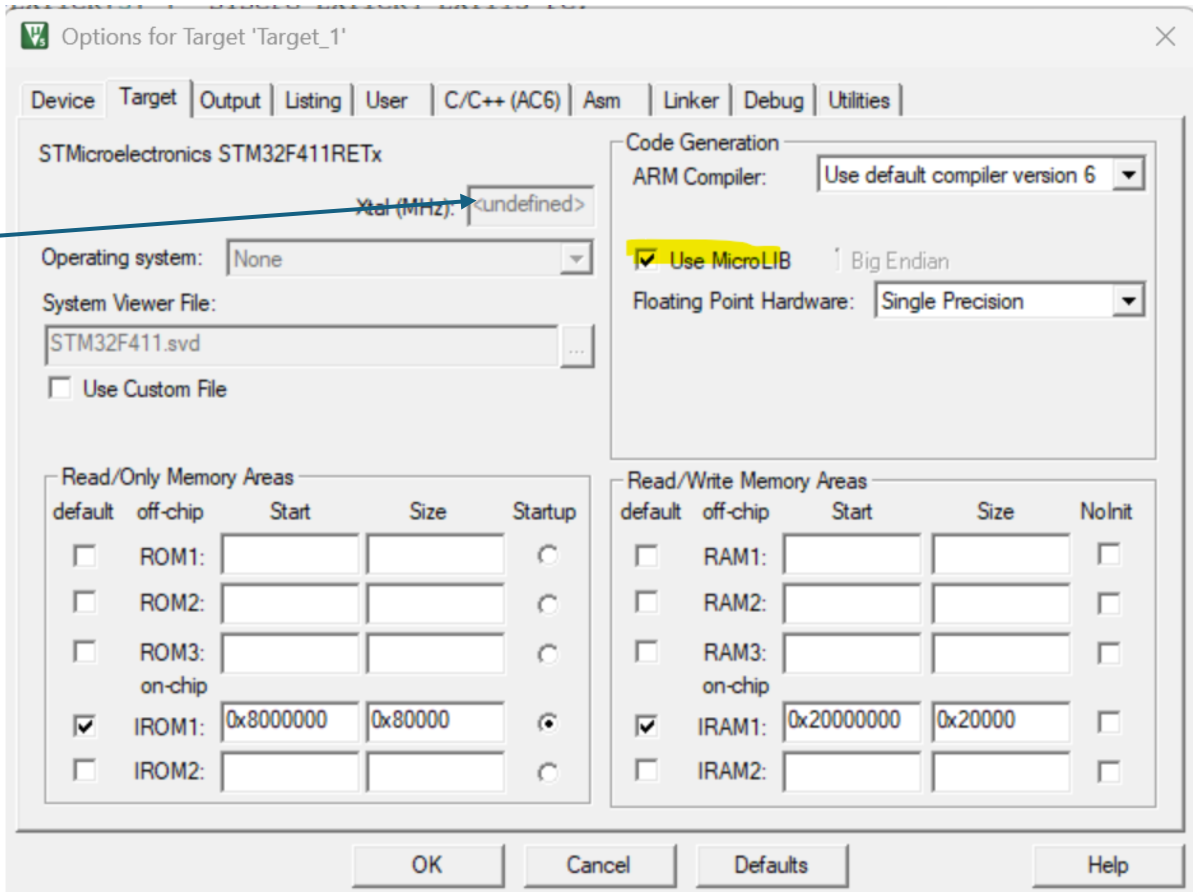


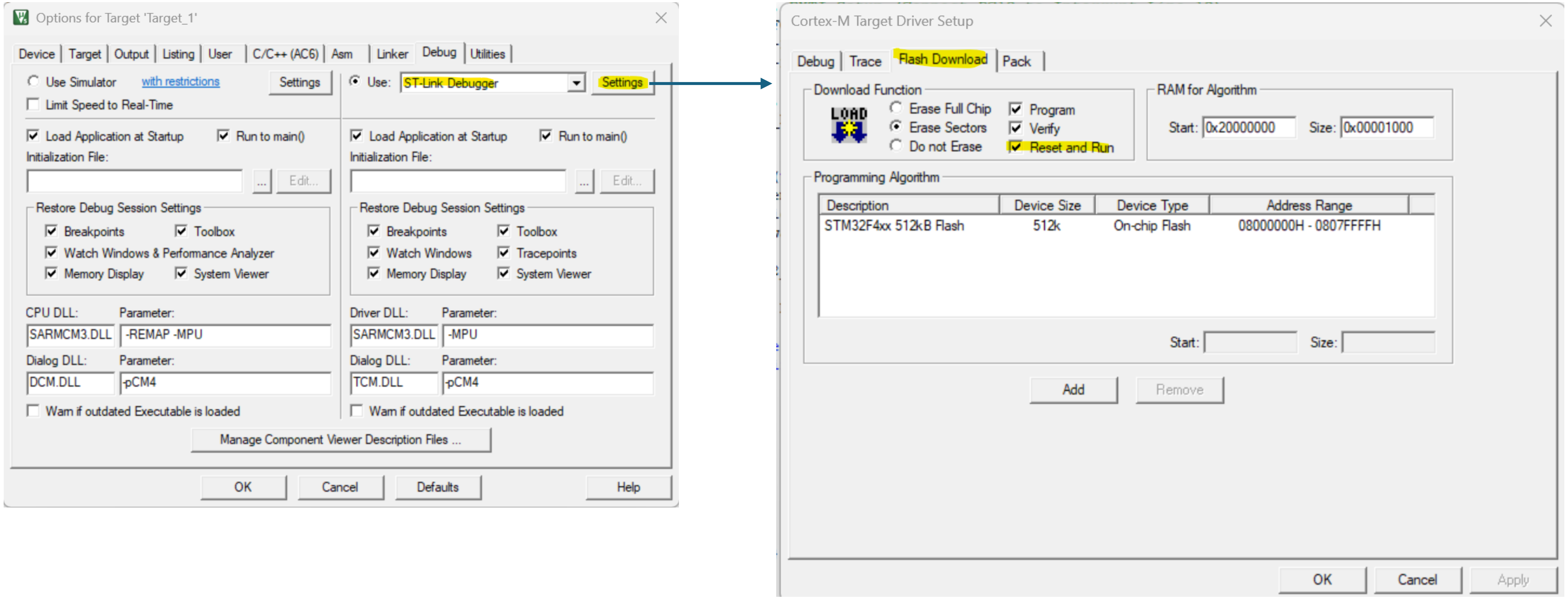
Phase 5: Configure Target Settings (The "Magic Wand")

Click the **Options for Target** icon (the magic wand).

1. **Target Tab:** * Set **Xtal (MHz)** to `16.0`.
   1. Check **Use MicroLIB** (required for `printf` and `sprintf`).
2. **Debug Tab:** * Ensure the right-hand side is set to **ST-Link Debugger**.
   1. Click **Settings** next to it, go to the **Flash Download** tab, and check **Reset and Run**.
3. Click **OK**.

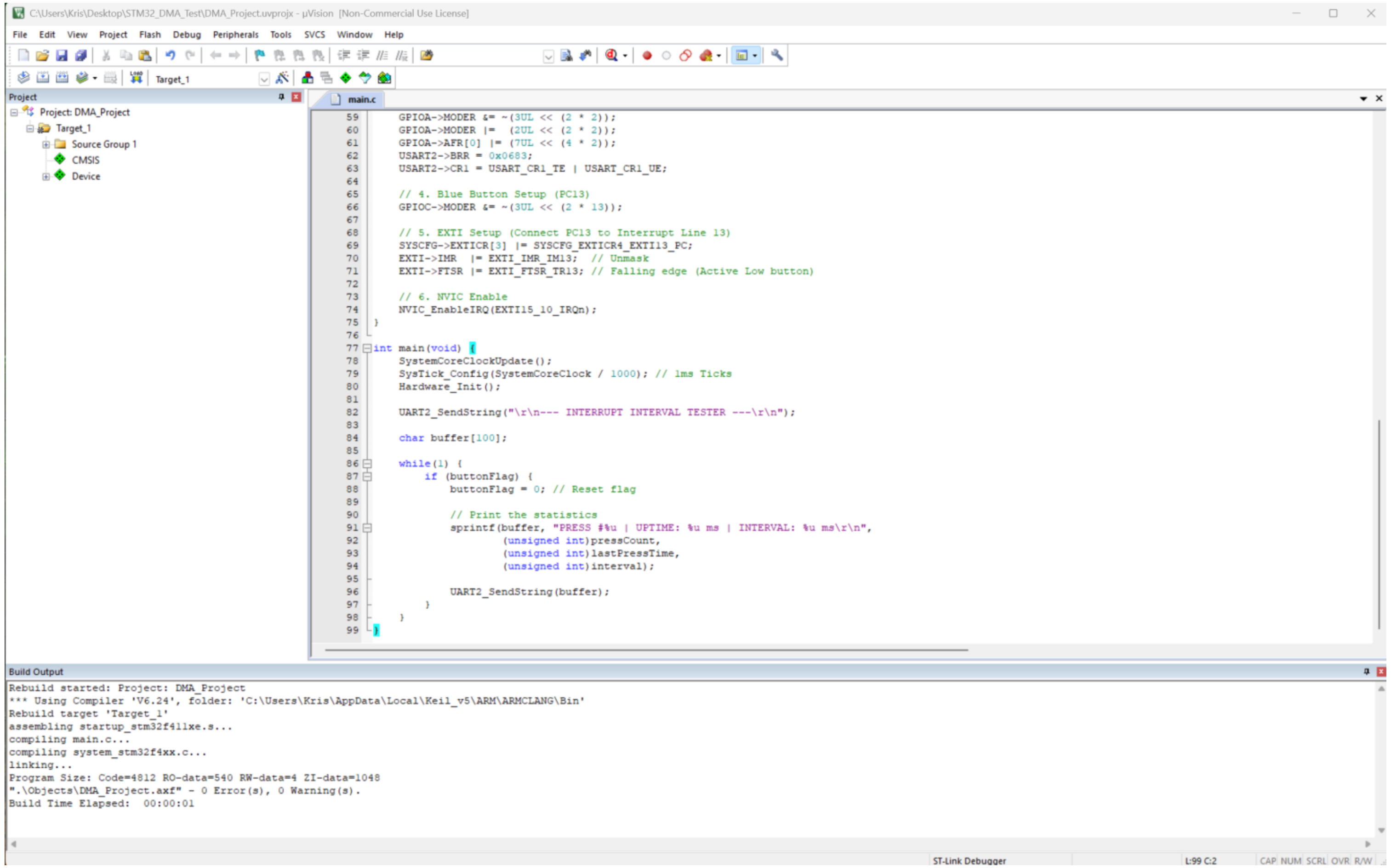
The reason **Xtal (MHz)** is greyed out in your **image_ef4942.png** is that you have successfully moved to the "Classic" 2.17.1 pack.
In the newer ARM Compiler 6 (AC6), Keil ignores that box because the clock speed is now defined physically inside your `system_stm32f4xx.c` file rather than a text box in the menu.

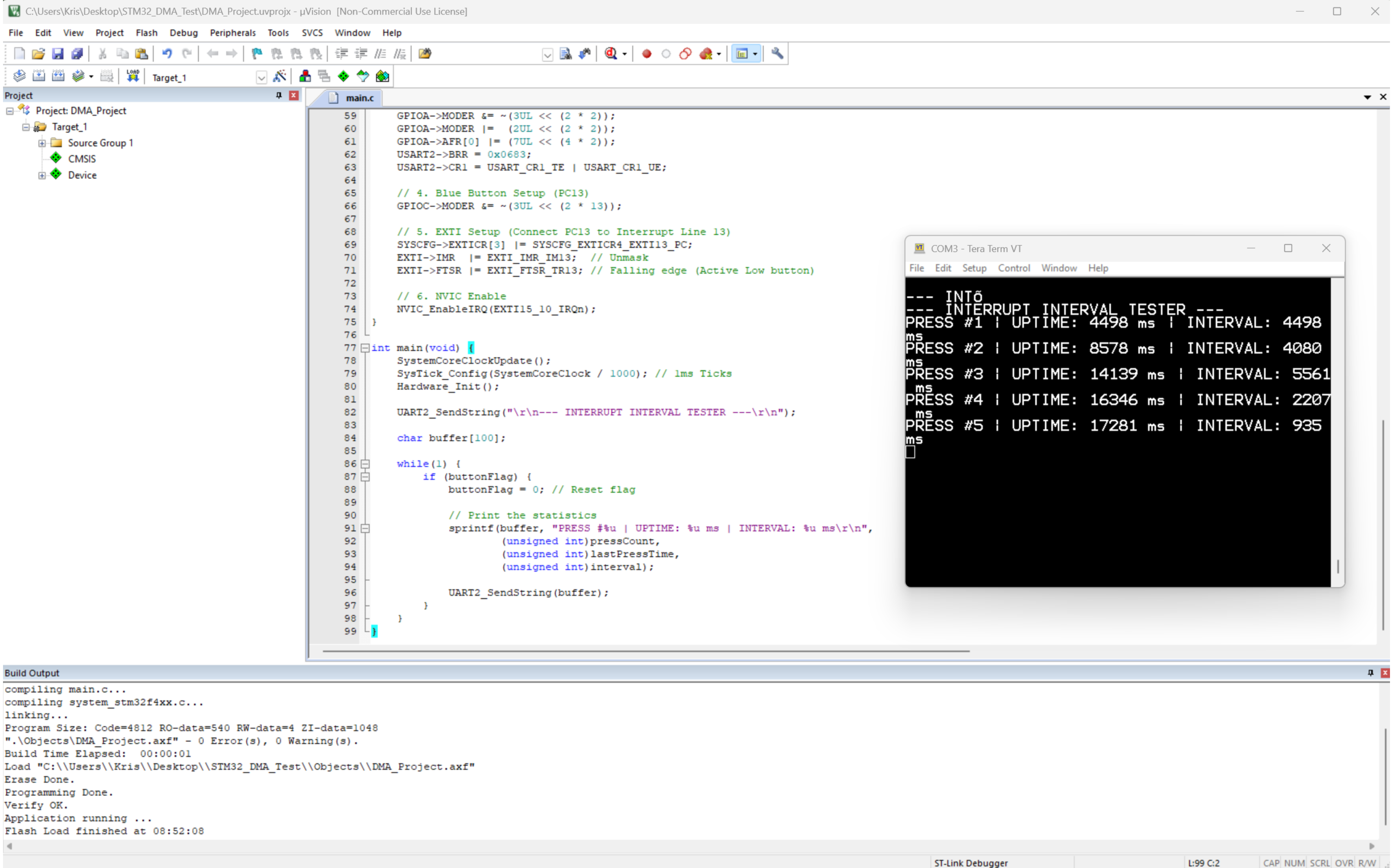



Options for Target 'Target_1'
Device | Target | Output | Listing | User | C/C++ (AC6) | Asm | Linker | Debug | Utilities
Use Simulator
with restrictions
Settings
Limit Speed to Real-Time
Use: ST-Link Debugger
Settings
Load Application at Startup
Run to main()
Initialization File:
Edit...
Restore Debug Session Settings
Breakpoints
Toolbox
Watch Windows & Performance Analyzer
Memory Display
System Viewer
Load Application at Startup
Run to main()
Initialization File:
Edit...
Restore Debug Session Settings
Breakpoints
Toolbox
Watch Windows
Tracepoints
Memory Display
System Viewer
CPU DLL:
Parameter:
SARMCM3.DLL
-REMAP -MPU
Dialog DLL:
Parameter:
DCM.DLL
-pCM4
Driver DLL:
Parameter:
SARMCM3.DLL
-MPU
Dialog DLL:
Parameter:
TCM.DLL
-pCM4
Warn if outdated Executable is loaded
Warn if outdated Executable is loaded
Manage Component Viewer Description Files ...
OK
Cancel
Defaults
Help
Cortex-M Target Driver Setup
Debug | Trace | Flash Download | Pack
Download Function
LOAD
Erase Full Chip
Erase Sectors
Do not Erase
Program
Verify
Reset and Run
RAM for Algorithm
Start: 0x20000000
Size: 0x00001000
Programming Algorithm
Description
Device Size
Device Type
Address Range
STM32F4xx 512kB Flash
512k
On-chip Flash
08000000H - 0807FFFFH
Start:
Size:
Add
Remove
OK
Cancel
Apply

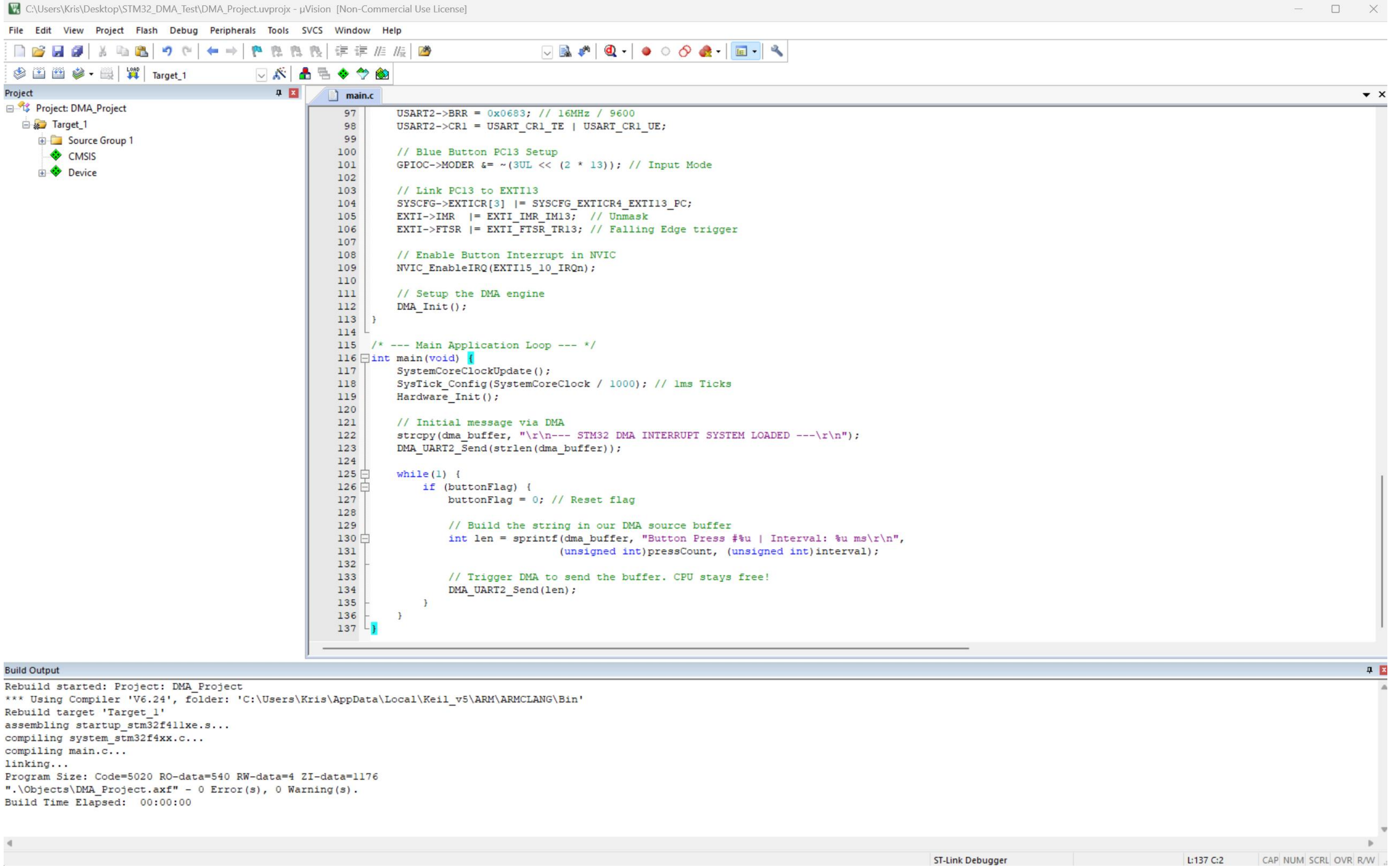
Phase 6: Build and Load

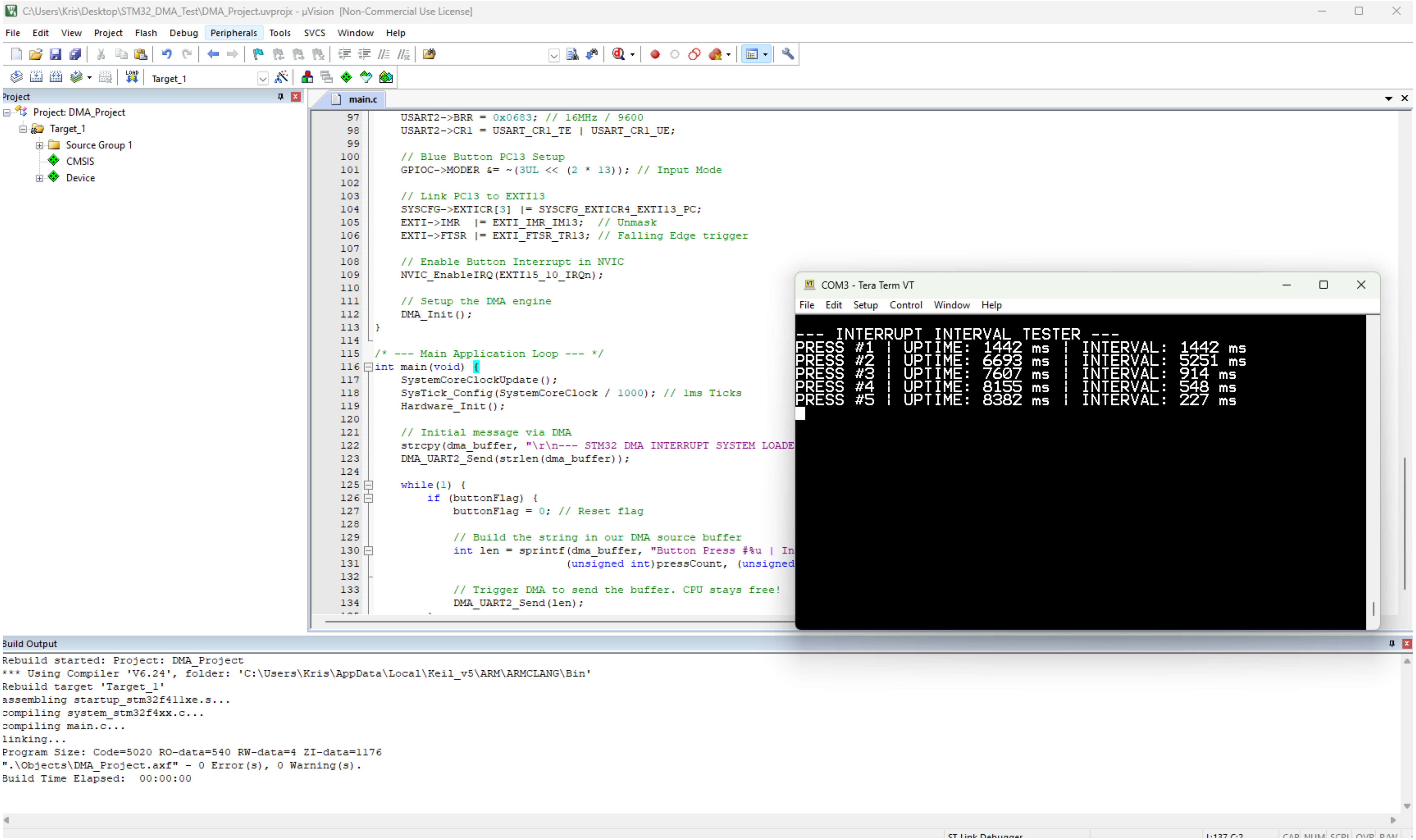
1. **Build:** Press **F7** (or the icon with two blue arrows). Check the bottom window for `0 Error(s)`.
2. **Load:** Press **F8** (or the **LOAD** icon) to send the code to your Nucleo board.
3. **Reset:** Press the **Black Button** on your board to start the program.

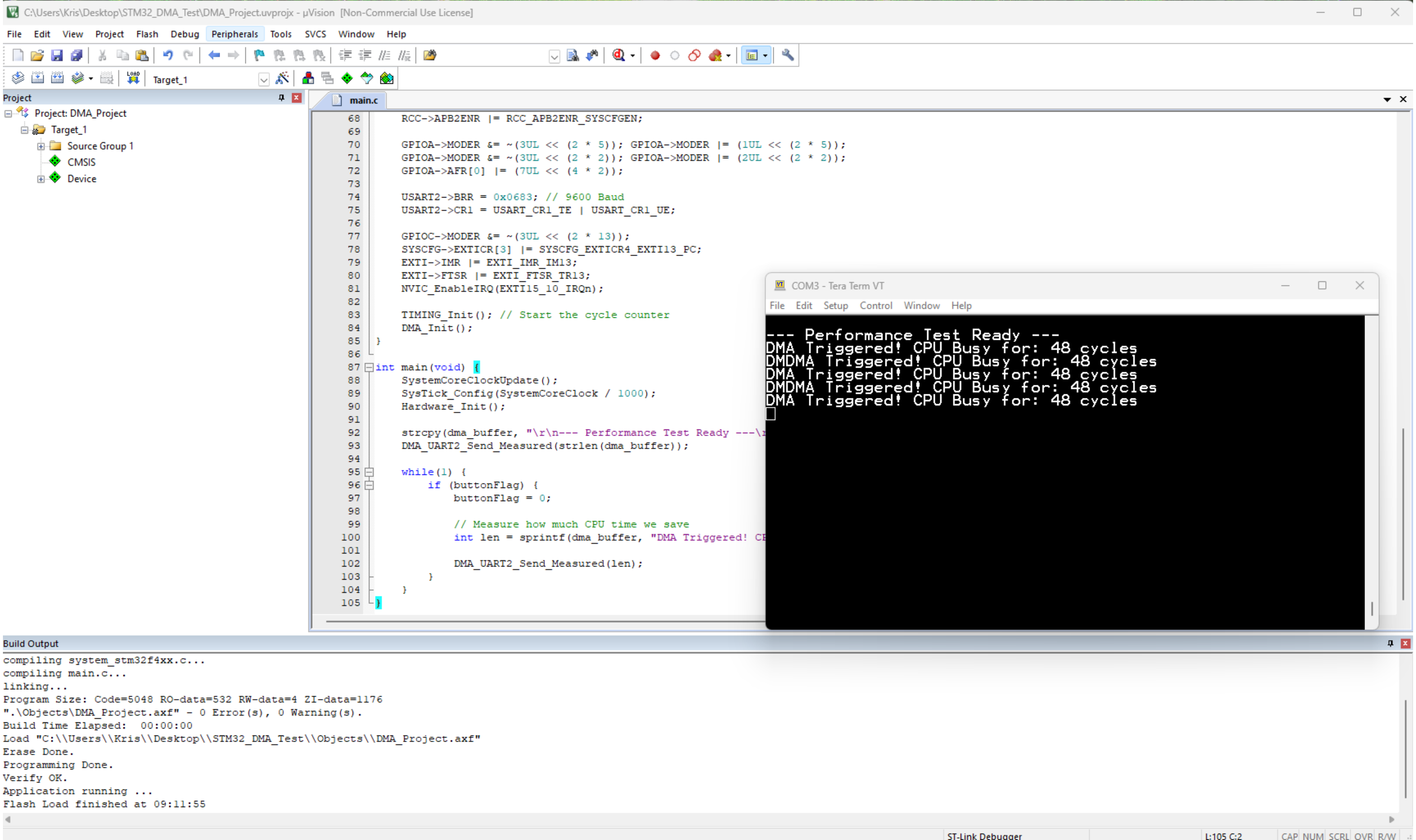


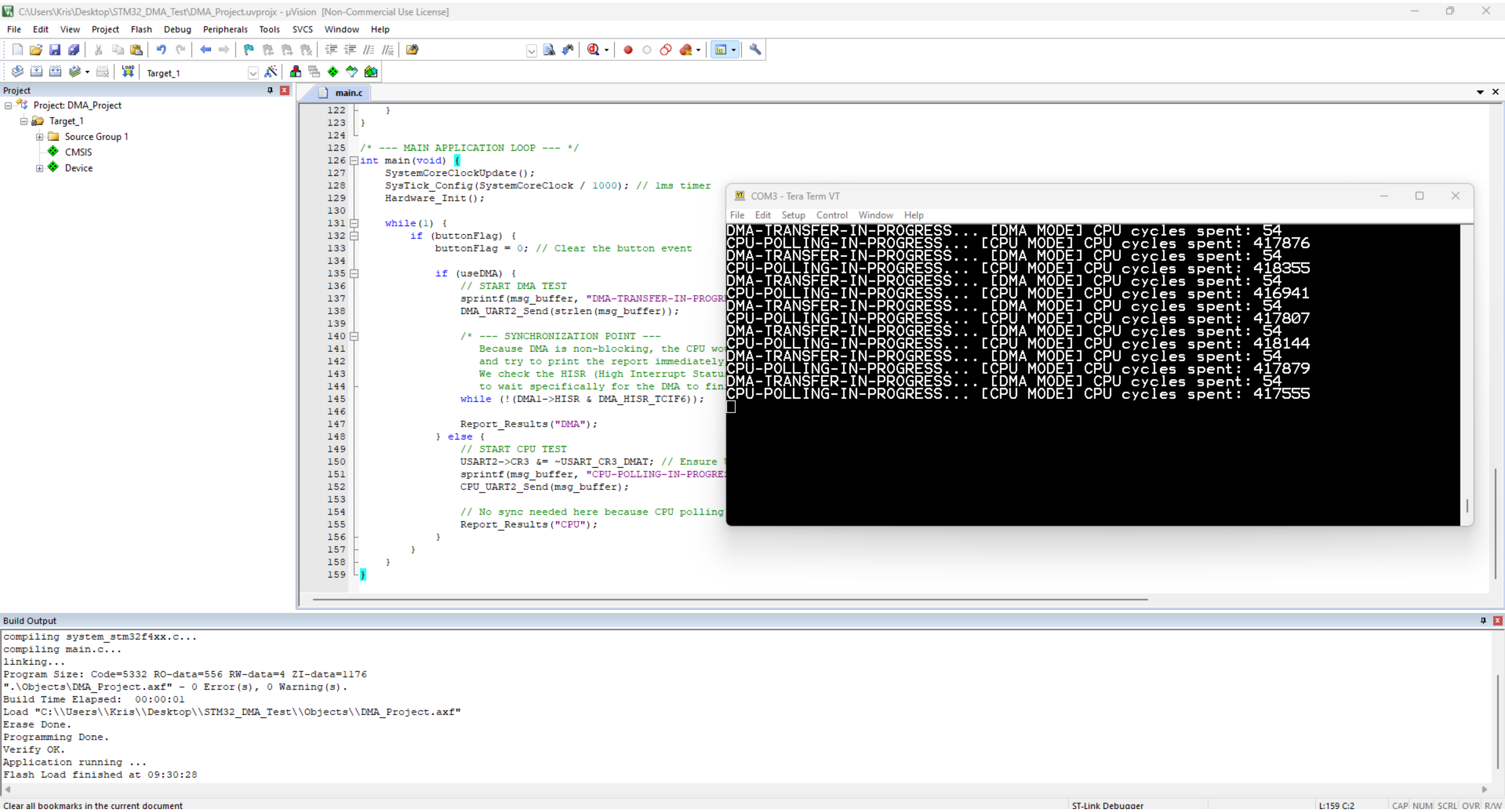

C:\Users\Kris\Desktop\STM32_DMA_Test\DMA_Project.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Target_1
Project
Project: DMA_Project
Target_1
Source Group 1
CMSIS
Device
main.c
GPIOA->MODER &= ~(3UL << (2 * 2));
GPIOA->MODER |= (2UL << (2 * 2));
GPIOA->AFR[0] |= (7UL << (4 * 2));
USART2->BRR = 0x0683;
USART2->CR1 = USART_CR1_TE | USART_CR1_UE;
// 4. Blue Button Setup (PC13)
GPIOC->MODER &= ~(3UL << (2 * 13));
// 5. EXTI Setup (Connect PC13 to Interrupt Line 13)
SYSCFG->EXTICR[3] |= SYSCFG_EXTICR4_EXTI13_PC;
EXTI->IMR |= EXTI_IMR_IM13; // Unmask
EXTI->FTSR |= EXTI_FTSR_TR13; // Falling edge (Active Low button)
// 6. NVIC Enable
NVIC_EnableIRQ(EXTI15_10_IRQn);
}
int main(void) {
SystemCoreClockUpdate();
SysTick_Config(SystemCoreClock / 1000); // 1ms Ticks
Hardware_Init();
UART2_SendString("\r\n--- INTERRUPT INTERVAL TESTER ---\r\n");
char buffer[100];
while(1) {
if (buttonFlag) {
buttonFlag = 0; // Reset flag
// Print the statistics
sprintf(buffer, "PRESS #%u | UPTIME: %u ms | INTERVAL: %u ms\r\n",
(unsigned int)pressCount,
(unsigned int)lastPressTime,
(unsigned int)interval);
UART2_SendString(buffer);
}
}
}
COM3 - Tera Term VT
File Edit Setup Control Window Help
--- INTõ
--- INTERRUPT INTERVAL TESTER ---
PRESS #1 | UPTIME: 4498 ms | INTERVAL: 4498 ms
PRESS #2 | UPTIME: 8578 ms | INTERVAL: 4080 ms
PRESS #3 | UPTIME: 14139 ms | INTERVAL: 5561 ms
PRESS #4 | UPTIME: 16346 ms | INTERVAL: 2207 ms
PRESS #5 | UPTIME: 17281 ms | INTERVAL: 935 ms
Build Output
compiling main.c...
compiling system_stm32f4xx.c...
linking...
Program Size: Code=4812 RO-data=540 RW-data=4 ZI-data=1048
".\Objects\DMA_Project.axf" - 0 Error(s), 0 Warning(s).
Build Time Elapsed: 00:00:01
Load "C:\\Users\\Kris\\Desktop\\STM32_DMA_Test\\Objects\\DMA_Project.axf"
Erase Done.
Programming Done.
Verify OK.
Application running ...
Flash Load finished at 08:52:08
ST-Link Debugger
L:99 C:2
CAP NUM SCRL OVR R/W

DMA Version of Code


C:\Users\Kris\Desktop\STM32_DMA_Test\DMA_Project.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Target_1
Project
Project: DMA_Project
Target_1
Source Group 1
CMSIS
Device
main.c
USART2->BRR = 0x0683; // 16MHz / 9600
USART2->CR1 = USART_CR1_TE | USART_CR1_UE;
// Blue Button PC13 Setup
GPIOC->MODER &= ~(3UL << (2 * 13)); // Input Mode
// Link PC13 to EXTI13
SYSCFG->EXTICR[3] |= SYSCFG_EXTICR4_EXTI13_PC;
EXTI->IMR |= EXTI_IMR_IM13; // Unmask
EXTI->FTSR |= EXTI_FTSR_TR13; // Falling Edge trigger
// Enable Button Interrupt in NVIC
NVIC_EnableIRQ(EXTI15_10_IRQn);
// Setup the DMA engine
DMA_Init();
}
/* --- Main Application Loop --- */
int main(void) {
SystemCoreClockUpdate();
SysTick_Config(SystemCoreClock / 1000); // 1ms Ticks
Hardware_Init();
// Initial message via DMA
strcpy(dma_buffer, "\r\n--- STM32 DMA INTERRUPT SYSTEM LOADED ---\r\n");
DMA_UART2_Send(strlen(dma_buffer));
while(1) {
if (buttonFlag) {
buttonFlag = 0; // Reset flag
// Build the string in our DMA source buffer
int len = sprintf(dma_buffer, "Button Press #%u | Interval: %u ms\r\n",
(unsigned int)pressCount, (unsigned int)interval);
// Trigger DMA to send the buffer. CPU stays free!
DMA_UART2_Send(len);
}
}
}
Build Output
Rebuild started: Project: DMA_Project
*** Using Compiler 'V6.24', folder: 'C:\Users\Kris\AppData\Local\Keil_v5\ARM\ARMCLANG\Bin'
Rebuild target 'Target_1'
assembling startup_stm32f411xe.s...
compiling system_stm32f4xx.c...
compiling main.c...
linking...
Program Size: Code=5020 RO-data=540 RW-data=4 ZI-data=1176
".\Objects\DMA_Project.axf" - 0 Error(s), 0 Warning(s).
Build Time Elapsed: 00:00:00
ST-Link Debugger
L:137 C:2
CAP NUM SCRL OVR R/W


C:\Users\Kris\Desktop\STM32_DMA_Test\DMA_Project.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Target_1
Project
Project: DMA_Project
Target_1
Source Group 1
CMSIS
Device
main.c
USART2->BRR = 0x0683; // 16MHz / 9600
USART2->CR1 = USART_CR1_TE | USART_CR1_UE;
// Blue Button PC13 Setup
GPIOC->MODER &= ~(3UL << (2 * 13)); // Input Mode
// Link PC13 to EXTI13
SYSCFG->EXTICR[3] |= SYSCFG_EXTICR4_EXTI13_PC;
EXTI->IMR |= EXTI_IMR_IM13; // Unmask
EXTI->FTSR |= EXTI_FTSR_TR13; // Falling Edge trigger
// Enable Button Interrupt in NVIC
NVIC_EnableIRQ(EXTI15_10_IRQn);
// Setup the DMA engine
DMA_Init();
}
/* --- Main Application Loop --- */
int main(void) {
SystemCoreClockUpdate();
SysTick_Config(SystemCoreClock / 1000); // 1ms Ticks
Hardware_Init();
// Initial message via DMA
strcpy(dma_buffer, "\r\n--- STM32 DMA INTERRUPT SYSTEM LOADE
DMA_UART2_Send(strlen(dma_buffer));
while(1) {
if (buttonFlag) {
buttonFlag = 0; // Reset flag
// Build the string in our DMA source buffer
int len = sprintf(dma_buffer, "Button Press #%u | In
(unsigned int)pressCount, (unsigned
// Trigger DMA to send the buffer. CPU stays free!
DMA_UART2_Send(len);
COM3 - Tera Term VT
File Edit Setup Control Window Help
--- INTERRUPT INTERVAL TESTER ---
PRESS #1 | UPTIME: 1442 ms | INTERVAL: 1442 ms
PRESS #2 | UPTIME: 6693 ms | INTERVAL: 5251 ms
PRESS #3 | UPTIME: 7607 ms | INTERVAL: 914 ms
PRESS #4 | UPTIME: 8155 ms | INTERVAL: 548 ms
PRESS #5 | UPTIME: 8382 ms | INTERVAL: 227 ms
Build Output
Rebuild started: Project: DMA_Project
*** Using Compiler 'V6.24', folder: 'C:\Users\Kris\AppData\Local\Keil_v5\ARM\ARMCLANG\Bin'
Rebuild target 'Target_1'
assembling startup_stm32f411xe.s...
compiling system_stm32f4xx.c...
compiling main.c...
linking...
Program Size: Code=5020 RO-data=540 RW-data=4 ZI-data=1176
".\Objects\DMA_Project.axf" - 0 Error(s), 0 Warning(s).
Build Time Elapsed: 00:00:00
ST-Link Debugger


C:\Users\Kris\Desktop\STM32_DMA_Test\DMA_Project.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Target_1
Project
Project: DMA_Project
Target_1
Source Group 1
CMSIS
Device
main.c
RCC->APB2ENR |= RCC_APB2ENR_SYSCFGEN;
GPIOA->MODER &= ~(3UL << (2 * 5)); GPIOA->MODER |= (1UL << (2 * 5));
GPIOA->MODER &= ~(3UL << (2 * 2)); GPIOA->MODER |= (2UL << (2 * 2));
GPIOA->AFR[0] |= (7UL << (4 * 2));
USART2->BRR = 0x0683; // 9600 Baud
USART2->CR1 = USART_CR1_TE | USART_CR1_UE;
GPIOC->MODER &= ~(3UL << (2 * 13));
SYSCFG->EXTICR[3] |= SYSCFG_EXTICR4_EXTI13_PC;
EXTI->IMR |= EXTI_IMR_IM13;
EXTI->FTSR |= EXTI_FTSR_TR13;
NVIC_EnableIRQ(EXTI15_10_IRQn);
TIMING_Init(); // Start the cycle counter
DMA_Init();
}
int main(void) {
SystemCoreClockUpdate();
SysTick_Config(SystemCoreClock / 1000);
Hardware_Init();
strcpy(dma_buffer, "\r\n--- Performance Test Ready ---\r
DMA_UART2_Send_Measured(strlen(dma_buffer));
while(1) {
if (buttonFlag) {
buttonFlag = 0;
// Measure how much CPU time we save
int len = sprintf(dma_buffer, "DMA Triggered! CP
DMA_UART2_Send_Measured(len);
}
}
}
COM3 - Tera Term VT
File Edit Setup Control Window Help
--- Performance Test Ready ---
DMA Triggered! CPU Busy for: 48 cycles
DMDMA Triggered! CPU Busy for: 48 cycles
DMA Triggered! CPU Busy for: 48 cycles
DMDMA Triggered! CPU Busy for: 48 cycles
DMA Triggered! CPU Busy for: 48 cycles
Build Output
compiling system_stm32f4xx.c...
compiling main.c...
linking...
Program Size: Code=5048 RO-data=532 RW-data=4 ZI-data=1176
".\Objects\DMA_Project.axf" - 0 Error(s), 0 Warning(s).
Build Time Elapsed: 00:00:00
Load "C:\\Users\\Kris\\Desktop\\STM32_DMA_Test\\Objects\\DMA_Project.axf"
Erase Done.
Programming Done.
Verify OK.
Application running ...
Flash Load finished at 09:11:55
ST-Link Debugger
L:105 C:2
CAP NUM SCRL OVR R/W


C:\Users\Kris\Desktop\STM32_DMA_Test\DMA_Project.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Target_1
Project
Project: DMA_Project
Target_1
Source Group 1
CMSIS
Device
main.c
}
}
/* --- MAIN APPLICATION LOOP --- */
int main(void) {
SystemCoreClockUpdate();
SysTick_Config(SystemCoreClock / 1000); // 1ms timer
Hardware_Init();
while(1) {
if (buttonFlag) {
buttonFlag = 0; // Clear the button event
if (useDMA) {
// START DMA TEST
sprintf(msg_buffer, "DMA-TRANSFER-IN-PROGR
DMA_UART2_Send(strlen(msg_buffer));
/* --- SYNCHRONIZATION POINT ---
Because DMA is non-blocking, the CPU wo
and try to print the report immediately
We check the HISR (High Interrupt Statu
to wait specifically for the DMA to fin
while (!(DMA1->HISR & DMA_HISR_TCIF6));
Report_Results("DMA");
} else {
// START CPU TEST
USART2->CR3 &= ~USART_CR3_DMAT; // Ensure
sprintf(msg_buffer, "CPU-POLLING-IN-PROGRE
CPU_UART2_Send(msg_buffer);
// No sync needed here because CPU polling
Report_Results("CPU");
}
}
}
}
COM3 - Tera Term VT
File Edit Setup Control Window Help
DMA-TRANSFER-IN-PROGRESS... [DMA MODE] CPU cycles spent: 54
CPU-POLLING-IN-PROGRESS... [CPU MODE] CPU cycles spent: 417876
DMA-TRANSFER-IN-PROGRESS... [DMA MODE] CPU cycles spent: 54
CPU-POLLING-IN-PROGRESS... [CPU MODE] CPU cycles spent: 418355
DMA-TRANSFER-IN-PROGRESS... [DMA MODE] CPU cycles spent: 54
CPU-POLLING-IN-PROGRESS... [CPU MODE] CPU cycles spent: 416941
DMA-TRANSFER-IN-PROGRESS... [DMA MODE] CPU cycles spent: 54
CPU-POLLING-IN-PROGRESS... [CPU MODE] CPU cycles spent: 417807
DMA-TRANSFER-IN-PROGRESS... [DMA MODE] CPU cycles spent: 54
CPU-POLLING-IN-PROGRESS... [CPU MODE] CPU cycles spent: 418144
DMA-TRANSFER-IN-PROGRESS... [DMA MODE] CPU cycles spent: 54
CPU-POLLING-IN-PROGRESS... [CPU MODE] CPU cycles spent: 417879
DMA-TRANSFER-IN-PROGRESS... [DMA MODE] CPU cycles spent: 54
CPU-POLLING-IN-PROGRESS... [CPU MODE] CPU cycles spent: 417555
Build Output
compiling system_stm32f4xx.c...
compiling main.c...
linking...
Program Size: Code=5332 RO-data=556 RW-data=4 ZI-data=1176
".\Objects\DMA_Project.axf" - 0 Error(s), 0 Warning(s).
Build Time Elapsed: 00:00:01
Load "C:\\Users\\Kris\\Desktop\\STM32_DMA_Test\\Objects\\DMA_Project.axf"
Erase Done.
Programming Done.
Verify OK.
Application running ...
Flash Load finished at 09:30:28
Clear all bookmarks in the current document
ST-Link Debugger
L:159 C:2
CAP NUM SCRL OVR R/W